

CSC 315 : COMPUTER ARCHITECTURE AND ORGANIZATION I

Week One: Lesson One

Introduction:

- The concept of computer architecture and organization.
- Fundamental building blocks (Structure and Function).
- The Von Neumann Architecture.

1.1 Organization and Architecture

In describing computers, a distinction is often made between *computer architecture* and *computer organization*. Although it is difficult to give precise definitions for these terms, a consensus exists about the general areas covered by each. For example, (Vranesic and Thurber, 1980; Siewiorek, Bell and Newell, 1982; Bell, Mudge and McNamara 1978).

Computer architecture refers to those attributes of a system visible to a programmer or, put another way, those attributes that have a direct impact on the logical execution of a program.

A term that is often used interchangeably with computer architecture is **instruction set architecture (ISA)**. The ISA defines instruction formats, instruction opcodes, registers, instruction and data memory; the effect of executed instructions on the registers and memory; and an algorithm for controlling instruction execution. **Examples of architectural attributes** include the instruction set, the number of bits used to represent various data types (e.g., numbers, characters), I/O mechanisms, and techniques for addressing memory.

Computer organization refers to the operational units and their interconnections that realize the architectural specifications.

Organizational attributes include those hardware details transparent to the programmer, such as control signals; interfaces between the computer and peripherals; and the memory technology used.

For example, it is an architectural design issue whether a computer will have a multiply instruction. It is an organizational issue whether that instruction will be implemented by a special multiply unit or by a mechanism that makes repeated use of the add unit of the system.

The organizational decision may be based on the anticipated frequency of use of the multiply instruction, the relative speed of the two approaches, and the cost and physical size of a special multiply unit.

Historically, and still today, the distinction between architecture and organization has been an important one. Many computer manufacturers offer a family of computer models, all with the same architecture but with differences in organization. A prominent example of both these phenomena is the IBM System/370 architecture.

This architecture was first introduced in 1970 and with a few enhancements, has survived to this day as the architecture of IBM's mainframe product line.

In microcomputers, the relationship between architecture and organization is very close. Changes in technology not only influence organization but also result in the introduction of more powerful and more complex architectures. Generally, there is less of a requirement for generation-to-generation compatibility for these smaller machines.

A computer organization must be designed to implement a particular architectural specification, therefore, a thorough treatment of organization requires a detailed examination of architecture, as well.

1.2 Structure and Function

A computer is a complex system; contemporary computers contain millions of elementary electronic components. How, then, can one clearly describe them? The key is to recognize *the hierarchical nature* of most complex systems, including the computer.

A hierarchical system is a set of interrelated subsystems, each of the latter, in turn, hierarchical in structure until we reach some lowest level of elementary subsystem.

The hierarchical nature of Computers is essential to both their design and their description. At each level, the designer is concerned with ***structure and function***:

- **Structure:** The way in which the components are interrelated.
- **Function:** The operation of each individual component as part of the structure.

In terms of description, we have two choices: *starting at the bottom and building up to a complete description*, or *beginning with a top view and decomposing the system into its subparts*.

Evidence from a number of fields suggests that the topdown approach is the clearest and most effective (Weinberg, 1975).

Function

Both the structure and functioning of a computer are, in essence, simple. In general terms, there are only four basic functions that a computer can perform:

- **Data processing:** Data may take a wide variety of forms, and the range of processing requirements is broad. Example of processing operations are; counting, swapping. Adding, and comparisons etc.
- **Data storage:** Even if the computer is processing data on the fly (i.e., data come in and get processed, and the results go out immediately), the computer *must temporarily store at least those pieces of data that are being worked on at any given moment*. Thus, there is at least a short-term data storage function. Equally important, *the computer performs a long-term data storage function*. Files of data are stored on the computer for subsequent retrieval and update.
- **Data movement:** The computer's operating environment consists of devices that serve as either sources or destinations of data. *When data are received from or delivered to a device that is directly connected to the computer, the process is known as input/output (I/O), and the device is referred to as a peripheral.* *When data are moved over longer distances, to or from a remote device, the process is known as data communications.*
- **Control:** Within the computer, a control unit *manages the computer's resources and orchestrates the performance of its functional parts in response to instructions*.

Structure (The Internal structure of the Computer)

A traditional computer employed a single processor and a microprogrammed control unit.

simple single-processor computer, as shown in Figure 1.1 provides a hierarchical view of the internal structure of a traditional single-processor computer.

There are four main structural components:

- **Central processing unit (CPU):** Controls the operation of the computer and performs its data processing functions; often simply referred to as processor.
- **Main memory:** Stores data.

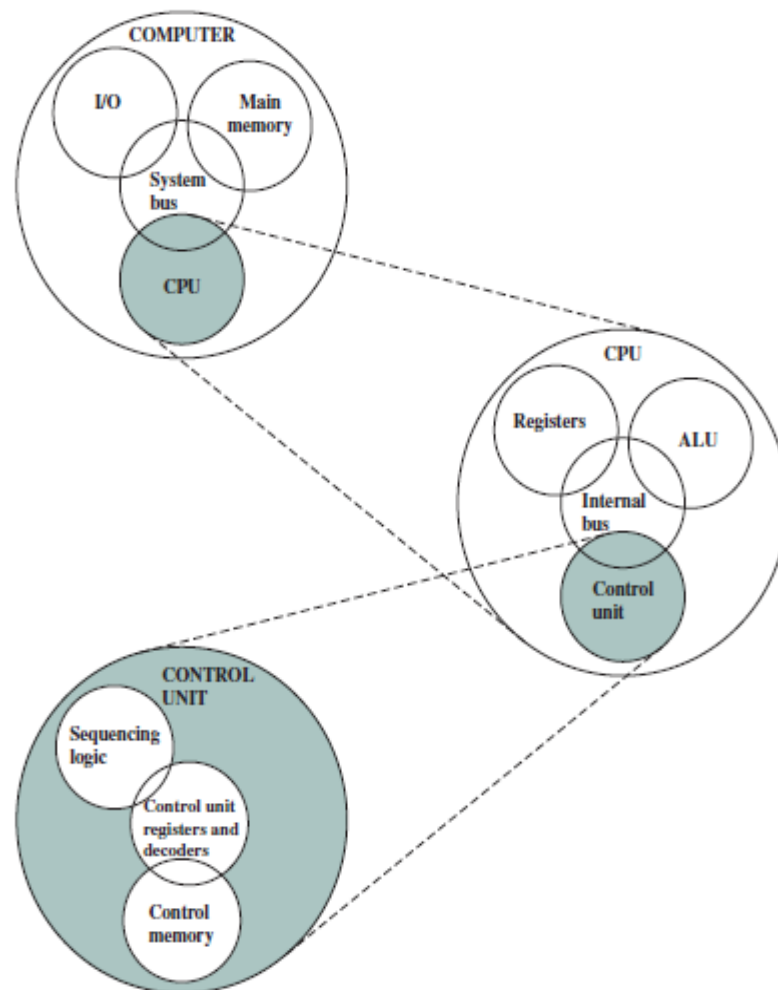


Figure 1.1: The Computer; top level structure

- **I/O:** Moves data between the computer and its external environment.
- **System interconnection:** Some mechanism that provides for communication among CPU, main memory, and I/O. A common example of system interconnection is by means of a **system bus**, consisting of a number of conducting wires to which all the other components attach.

There may be one or more of each of the aforementioned components. Traditionally, there has been just a single processor, however, In recent years, there has been increasing use of multiple processors in a single computer.

The most interesting and in some ways the most complex component is the CPU. Its major structural components are as follows:

- **Control unit:** Controls the operation of the CPU and hence the computer.
- **Arithmetic and logic unit (ALU):** Performs the computer's data processing functions.
- **Registers:** Provides storage internal to the CPU.
- **CPU interconnection:** Some mechanism that provides for communication among the control unit, ALU, and registers.

Complexity is added by the use of parallel and pipelined organizational techniques.

Finally, there are several approaches to the implementation of the control unit; one common approach is a *microprogrammed* implementation. In essence, a microprogrammed control unit operates by executing microinstructions that define the functionality of the control unit. With this approach, the structure of the control unit can be depicted, as in **Figure 1.1**.

multicore computer structure

Many contemporary computers generally have multiple processors. When these processors all reside on a single chip, the term ***multicore computer*** is used, and each processing unit (consisting of a control unit, ALU, registers, and perhaps cache) is called a *core*.

To clarify the terminology, this text will use the following definitions.

- **Central processing unit (CPU):** That portion of a computer that fetches and executes instructions. It consists of an ALU, a control unit, and registers. In a system with a single processing unit, it is often simply referred to as a *processor*.
- **Core:** An individual processing unit on a processor chip. A core may be equivalent in functionality to a CPU on a single CPU system. Other specialized processing units, such as one optimized for vector and matrix operations, are also referred to as cores.
- **Processor:** A physical piece of silicon containing one or more cores. The processor is the computer component that interprets and executes instructions. If a processor contains multiple cores, it is referred to as a multicore processor.

Figure 1.2 is a simplified view of the principal components of a typical multicore computer.

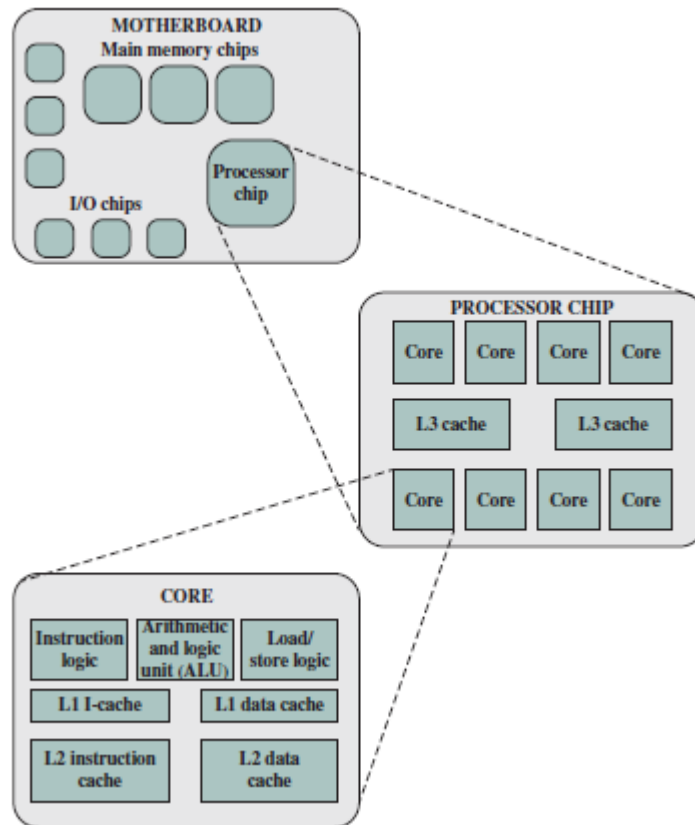


Figure 1.2: Simplified view of elements of a Multicore Computer

Most computers, including embedded computers in smartphones and tablets, plus personal computers, laptops, and workstations, are housed on a motherboard.

A printed circuit board (PCB) is a rigid, flat board that holds and interconnects chips and other electronic components. The board is made of layers, typically two to ten, that interconnect components via copper pathways that are etched into the board. The main printed circuit board in a computer is called a system board or motherboard, while smaller ones that plug into the slots in the main board are called expansion boards.

The most prominent elements on the motherboard are the chips. A chip is a single piece of semiconducting material, typically silicon, upon which electronic circuits and logic gates are fabricated. The resulting product is referred to as an integrated circuit.

The motherboard contains a slot or socket for the processor chip, which typically contains multiple individual cores, in what is known as a multicore processor.

There are also slots for memory chips, I/O controller chips, and other key computer components.

For desktop computers, expansion slots enable the inclusion of more components on expansion boards. Thus, a modern motherboard connects only a few individual chip components, with each chip containing from a few thousand up to hundreds of millions of transistors.

Figure 1.2 above shows a processor chip that contains eight cores and an L3 cache. Not shown is the logic required to control operations between the cores and the cache and between the cores and the external circuitry on the motherboard.

The figure indicates that the L3 cache occupies two distinct portions of the chip surface. However, typically, all cores have access to the entire L3 cache via the control circuits. The processor chip shown in Figure 1.2 does not represent any specific product, but provides a general idea of how such chips are laid out.

The structure of a single core, which occupies a portion of the processor chip.

In general terms, the functional elements of a core are:

- **Instruction logic:** This includes the tasks involved in fetching instructions, and decoding each instruction to determine the instruction operation and the memory locations of any operands.
- **Arithmetic and logic unit (ALU):** Performs the operation specified by an instruction.
- **Load/store logic:** Manages the transfer of data to and from main memory via cache.

The core also contains an L1 cache, split between an instruction cache (I-cache) that is used for the transfer of instructions to and from main memory, and an L1 data cache, for the transfer of operands and results. Typically, today's processor chips also include an L2 cache as part of the core.

In many cases, this cache is also split between instruction and data caches, although a combined, single L2 cache is also used.

Keep in mind that this representation of the layout of the core is only intended to give a general idea of internal core structure. In a given product, the functional

elements may not be laid out as the three distinct elements shown in Figure 1.2, especially if some or all of these functions are implemented as part of a microprogrammed control unit. Figure 1.3 is a photograph of the motherboard for a computer built around two Intel Quad-Core Xeon processor chips.

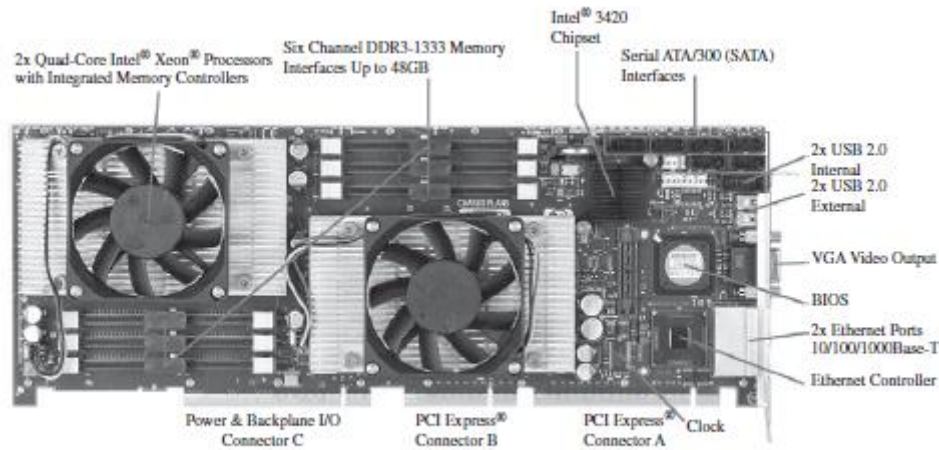


Figure 1.3 Motherboard with Two Intel Quad-Core Xeon Processors
Source: Chassis Plans, www.chassis-plans.com

Figure 1.3: MotherBoard with two Intel Quad-Core Xeon Processors

The most important in addition to the processor sockets:

- PCI-Express slots for a high-end display adapter and for additional peripherals .
- Ethernet controller and Ethernet ports for network connections.
- USB sockets for peripheral devices.
- Serial ATA (SATA) sockets for connection to disk memory Interfaces for DDR (double data rate) main memory chips
- Intel 3420 chipset is an I/O controller for direct memory access operations between peripheral devices and main memory

Read on the internal structure of a processor chip.

1.3 Von Neumann architecture

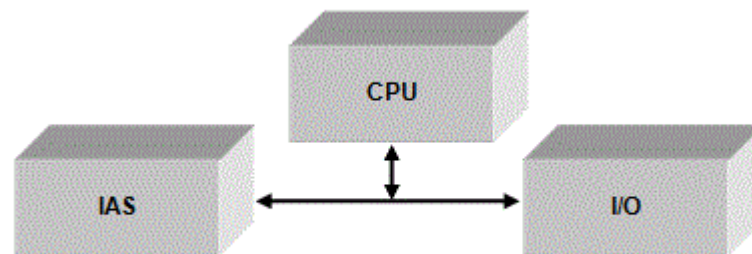
In the 1940s, a mathematician called John Von Neumann described the basic arrangement (or architecture) of a computer.

Most computers today follow the concept that he described although there are other types of architecture.

When we talk about the Von Neumann architecture, we are actually talking about the relationship between the hardware that makes up a Von Neumann-based computer.

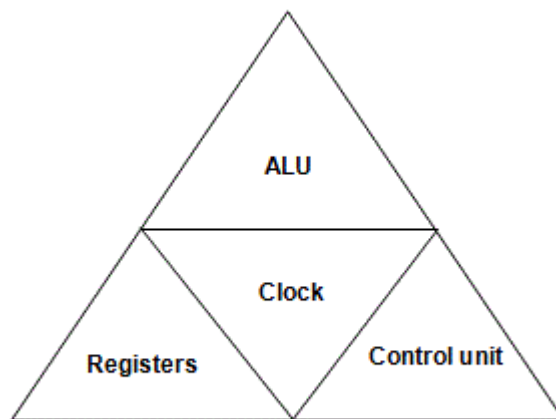
A Von Neumann-based computer is a computer that:

- Uses a single processor.
- Uses one memory (and single path) for both instructions and data. A von Neumann computer cannot distinguish between data and instructions in a memory location! It 'knows' only because of the location of a particular bit pattern in RAM.
- Executes programs by doing one instruction after the next in a serial manner using a fetch-decode-execute cycle.



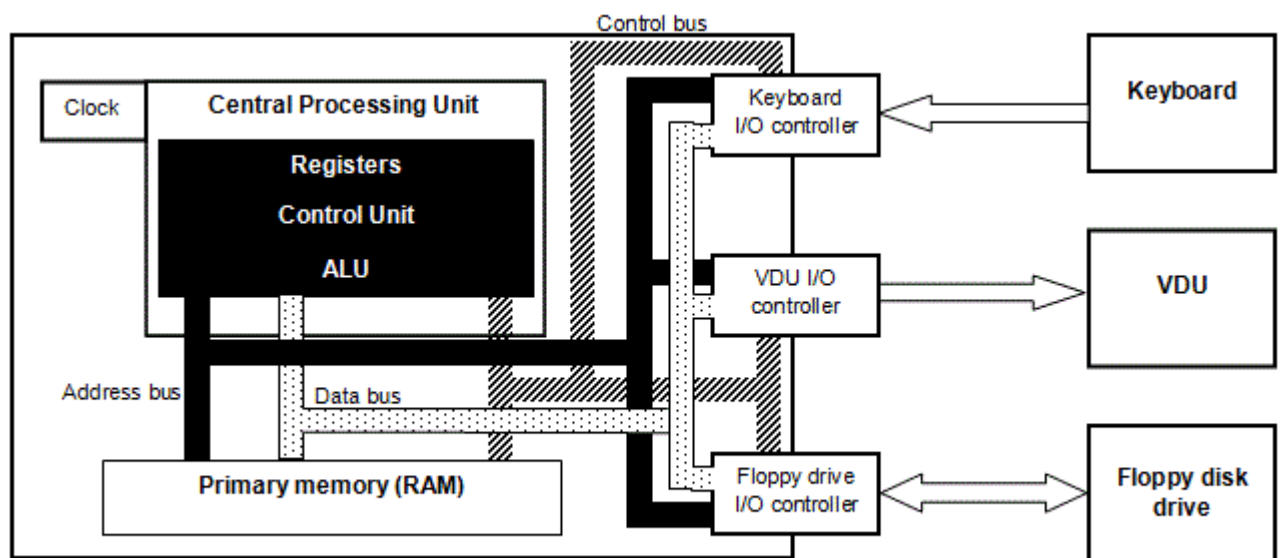
A model of a Von Neumann computer system.

Figure 1.4: A model of the Von Neumann Computer System



A diagrammatic representation of the 4 main parts of a CPU.

Figure 1.6: A diagrammatic representation of the 4 main parts of a CPU



A computer system showing the I/O controllers.

Figure 1.6: A computer system showing I/O controllers

Von Neumann bottleneck

Whatever you do to improve performance, you cannot get away from the fact that instructions can only be done one at a time and can only be carried out sequentially.

Both of these factors hold back the efficiency of the CPU. This is commonly referred to as the 'Von Neumann bottleneck'.

You can provide a Von Neumann processor with more RAM, more cache or faster components but if real gains are to be made in CPU performance then a major review needs to take place of CPU design.